



Fundamentals of Business Analytics

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TOPIC TO BE DISCUSS:

Module 7:

WHAT RESOURCE CONSIDERATIONS ARE
IMPORTANT TO SUPPORT BUSINESS
ANALYTICS?

Lesson 1:

BUSINESS ANALYTICS PERSONNEL



LESSON 1: BUSINESS ANALYTICS PERSONNEL

According to business.wfu.edu (2024), for professionals using AI, identifying patterns isn't quite as simple as applying a formula and reading the results. Instead, the process entails:

- ✓ **aggregating the data**, as more than one structured and unstructured source may be involved;
- ✓ **data mining**, or classifying the data and identifying trends;
- ✓ **forecasting**, or using the results of data mining to influence future business decisions; and
- ✓ **data visualization**, or the presentation of data mining and forecasting into an accessible format that non-technical professionals can understand.



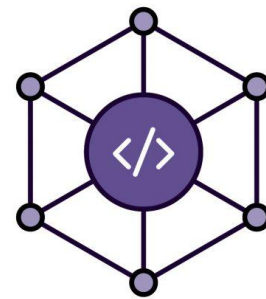
LESSON 1: BUSINESS ANALYTICS PERSONNEL

*The **business analytics professional**—sometimes called a **data analyst**—then approaches the data from a higher level, examining it for patterns, making it usable, creating models from it, and finding ways in which these numbers and strings of text can improve department or business processes or make future financial or performance projections (business.wfu.edu, 2024).*

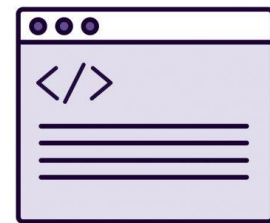


LESSON 1: BUSINESS ANALYTICS PERSONNEL

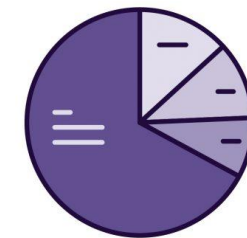
7 Essential Data Analyst Skills



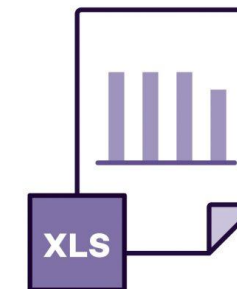
SQL and
NoSQL



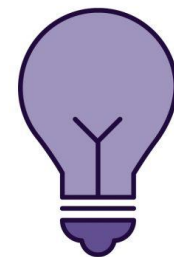
R, Python,
and MATLAB



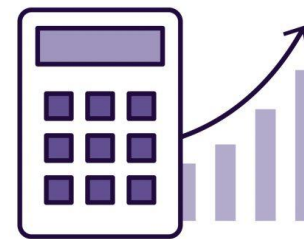
Data
Visualization



Microsoft
Excel



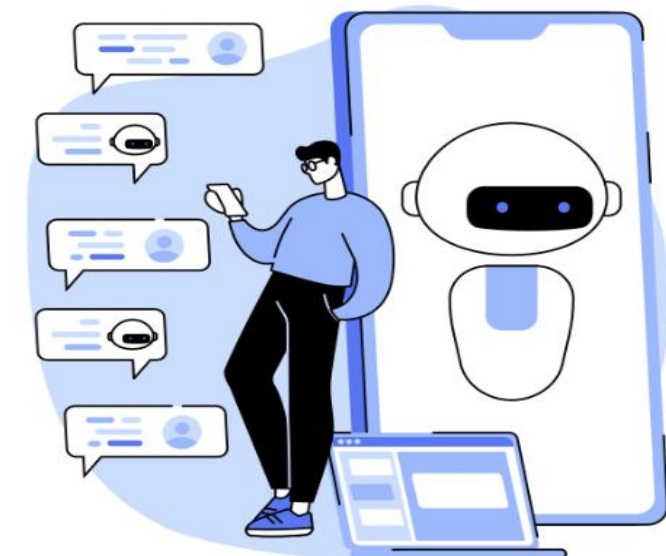
Critical
Thinking



Math and
Statistics



Communication



LESSON 1: BUSINESS ANALYTICS PERSONNEL

➤ Roles and responsibilities of a business analyst

Some initiatives they might provide analysis for include the following:

- ✓ Identifying strategic opportunities from data patterns.
- ✓ Identifying potential problems facing the business and coming up with solutions.
- ✓ Creating a budget and business forecast.
- ✓ Monitoring progress with business initiatives.
- ✓ Reporting progress on business objectives back to stakeholders.
- ✓ Understanding KPIs.
- ✓ Understanding regulatory and reporting requirements.



LESSON 1: BUSINESS ANALYTICS PERSONNEL

Business analysts must have a mixture of **hard skills** and **soft skills**. A business analyst *doesn't* need a deep understanding of IT but does need to understand how systems work together.

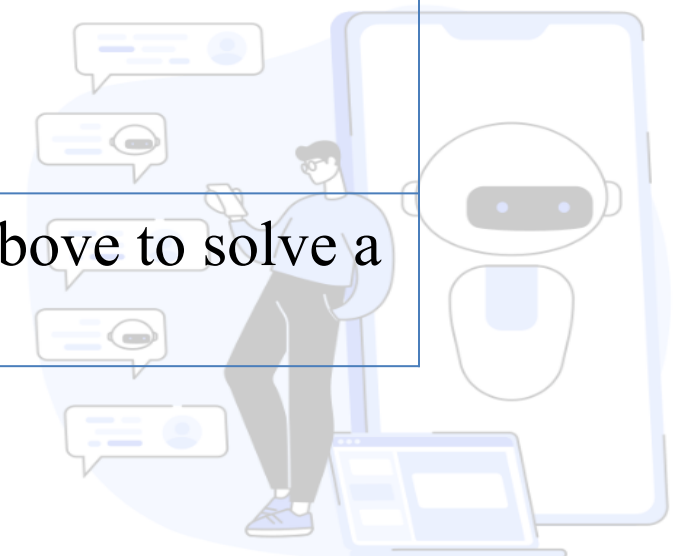
The **data scientist**, meanwhile, is the senior-most member of this group. This professional often acts like an investigator, searching for ways business processes could be improved and then backing up any proposals or solutions with data.



LESSON 1: BUSINESS ANALYTICS PERSONNEL

Table 7.1 Cognizure Organization Certification Exam Content Areas

Exam	Topic	Specific Content Areas Covered	Example
I	Statistical Methods	1. Visualizing and Exploring Data 2. Descriptive Statistics 3. Probability Distributions 4. Sampling and Estimation 5. Statistical Inference 6. Regression Analysis 7. Predictive Modeling and Analysis	1. Graphs and Charts 2. Mean, median, mode 3. Normal Distribution 4. Confidence intervals 5. Hypothesis Testing 6. Multiple regression 7. Curve fitting of models and functions to raw data
II	Operations Research Methods	1. Linear Optimization 2. Integer Optimization 3. Nonlinear Optimization 4. Simulation 5. Decision Analysis 6. Forecasting	1. Linear programming 2. Integer programming 3. Quadratic programming 4. Monte Carlo method 5. Expected value analysis 6. Exponential smoothing
III	Case Studies	Practical knowledge of real-world situations	Application of the methods above to solve a real world problem



LESSON 1: BUSINESS ANALYTICS PERSONNEL

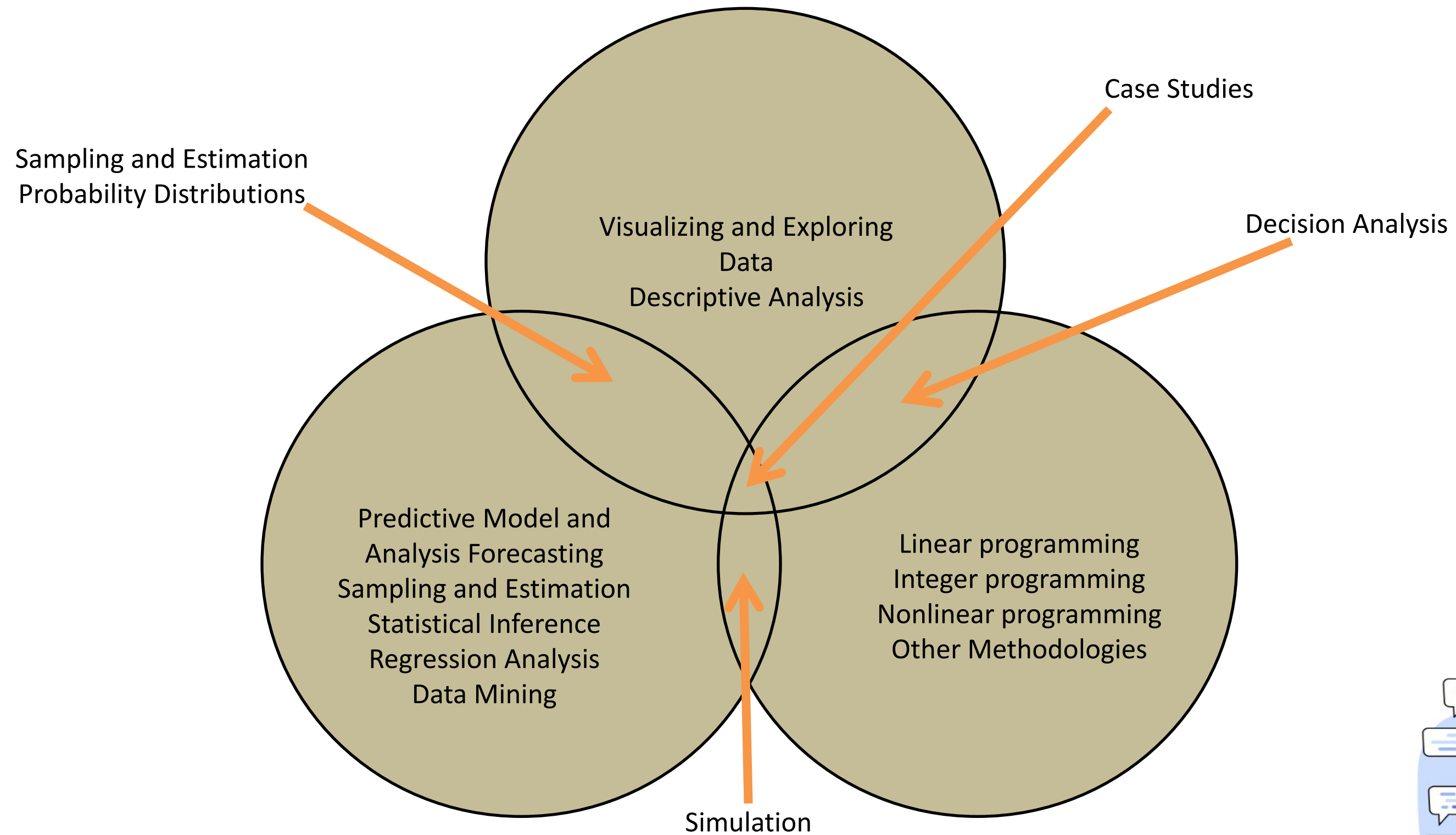


Figure 7.1 Certification content areas and their relationship to the steps in BA

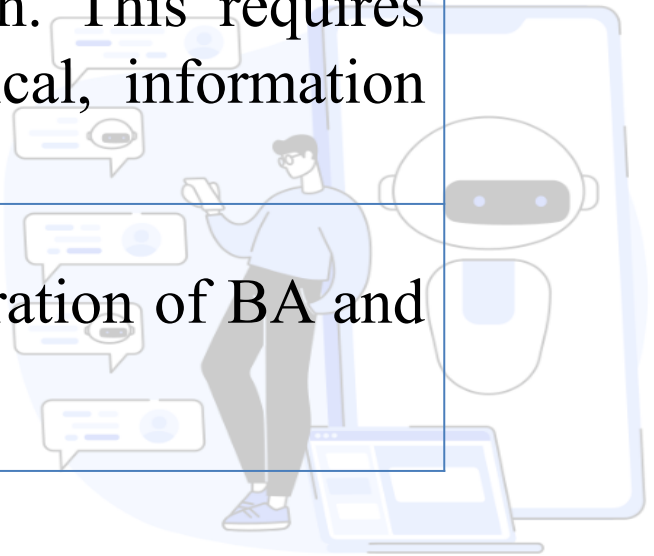
Source: Marc J. Schniederjans, et. al. (2020)



LESSON 1: BUSINESS ANALYTICS PERSONNEL

Table 7.2 Types of BA Personnel

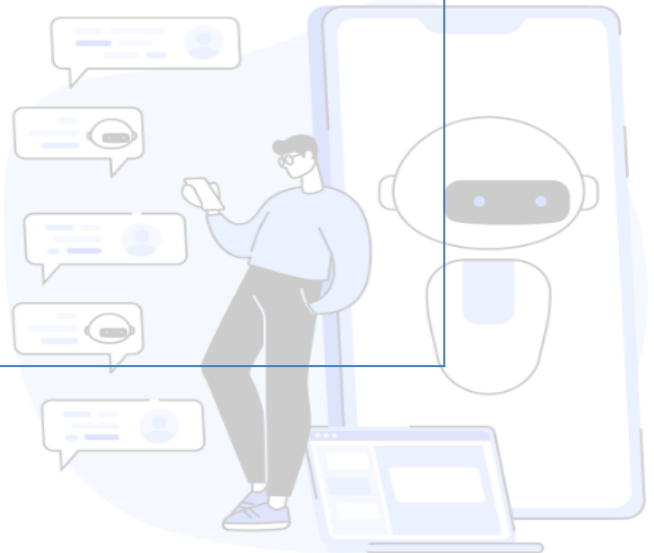
BA Personnel Specialty	Description
Administrators	Within the context of the IBM BA and business intelligence (BI) software platforms, administrators manage servers (their load balancing, installation, and configurations). They manage reports from computer portals, manage dispatchers, and perform troubleshooting for technology. They are also in charge of user authorization and authentication for security.
Designers	As members of a team, designers are responsible for building reports using relational data models, as well as enhancing, customizing, and managing professional reports.
Developers	As members of a team, the skills required for developers are closely tied to the BA process and involve the application of analytics, data warehousing, model building, use of operations research and statistical methodologies, and real-time monitoring of data flow to users.
Solution Experts	As members of a team, solution experts analyze, plan, design, deploy, and operate BA applications using an appropriate methodology and development approach. This requires knowledge in many differing BA software applications, including statistical, information systems, and operations research methods.
Technical Specialists	As members of a team, they are responsible for the installation and configuration of BA and BI applications.



LESSON 1: BUSINESS ANALYTICS PERSONNEL

Table 7.3 Select Types of BA Personnel Skills or Competency Requirements

Type of Skill or Competency	Description of Possible Roles
Business	• Leadership • People-related management and communication skills • Manage BA projects (prioritize, schedule, and so on) • Manage BA processes (rules, procedures governing acceptance and use) • Determine project requirements • Train BA personnel to develop competencies
Analytics	• Know how to use statistical methodologies • Know how to use operations research methodologies • Know how to use data mining for quantitative data and text analytics for unstructured data
Information system	• Maintain and use computer portals • Identify and extract data • Maintain quality data



LESSON 1: BUSINESS ANALYTICS PERSONNEL

In summary, people who undertake a career in BA are ***expected*** to know how to **interact with people** and utilize the necessary analytic tools to leverage data into useful information that can be processed, stored, and shared in information systems in a way that guides a firm to higher levels of business performance (Marc J. Schniederjans, et.al., 2020).



TOPICS TO BE DISCUSS:

Module 7

Lesson 2 : **Business Analytics Data**

Lesson 2.1: **Categorizing Data**

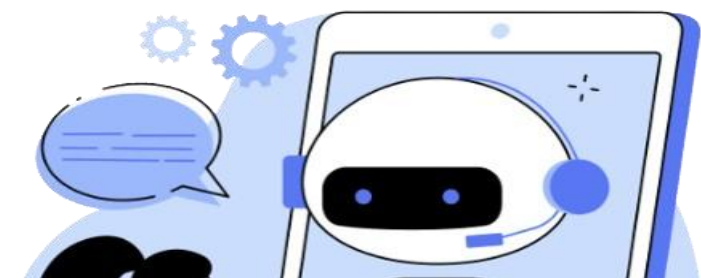
Lesson 2.2: **Data Issue**



LESSON 2: Business Analytics Data

Data in Business Analytics

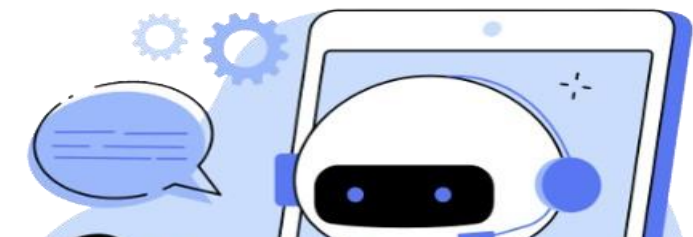
In the world of business analytics, data is the foundation of everything. Data in business analytics refers to the information that is collected, processed, and analyzed to gain insights and make informed decisions. Data can come from a variety of sources, such as customer interactions, website traffic, financial transactions, and social media activity.



Types of Data

Structured Data is a type of data that is organized and easily managed using traditional data management tools such as spreadsheets, databases, or tables. Structured data is typically quantitative and numeric in nature, meaning that it consist of numbers, percentages, and other numerical values.

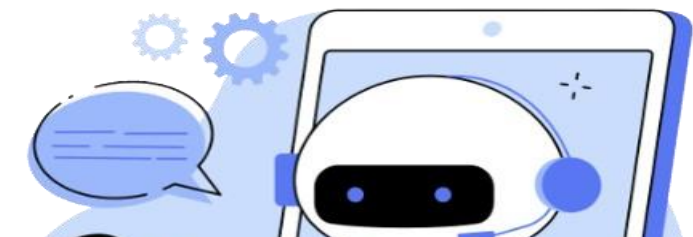
Unstructured Data is data that does not have a predefined format or organization, making it difficult to manage using traditional data management tools. Examples of unstructured data include social media posts, emails, images, and videos.



Types of Data

Semi-Structured Data is a type of data that has elements of both structured and unstructured data. This type of data includes information that is partially organized, but not to the extent that it can be classified as structured data

Big Data is a term used to describe large and complex data sets that cannot be processed using traditional data management tools. Big Data includes a variety of data types, including structured, unstructured, and semi-structured data. The main challenge of analyzing Big Data is its volume, as the amount of data is too large to be analyzed manually.



LESSON 2.1: Categorizing Data

There are many ways to categorize business analytics data. Data is commonly categorized by either **internal** or **external** sources.

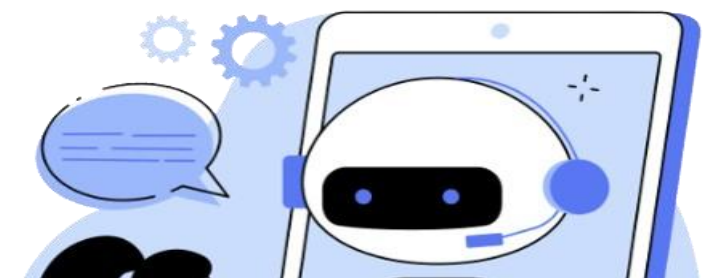


Table 7.4 Typical Internal Sources of Data on Which Business Analytics Can Be Based

Type of Internal Data	Description
Billing and Reminder Systems	Billing systems and reminder systems print bills and monitor customer payment information on valued-based customer segments.
Business	Industry codes, accounting information, personnel information, and so on are routinely collected in the course of business.
Customer	Names, addresses, returns, special contracts, segmentations, and so on are obtained when customers sign for or pay for products or services.
Customer Relationship Management Systems	<i>Customer relationship management</i> (CRM) systems collect and provide data on customer history, behavior on matters like complaints, the end of a relationship with a firm, and so on.
Human Resources	Information about employees, salaries, competencies, and so on is recorded by routine efforts over the history of employment.
Information from Enterprise Resource Planning Systems	<i>Enterprise resource planning</i> (ERP) systems are used to communicate internal business transactions to provide a direct feed of information on management issues and concerns, as well as other operations activities required to produce and sell products.
Product	Information is collected from procurement through post sales to monitor profitability, durability, and quality.
Production	Information that can be used to optimize production, inventory control, and supply chain delivery of the product to the customers is collected during the production processes.
Questionnaires	Information on customer behavior is obtained by customer

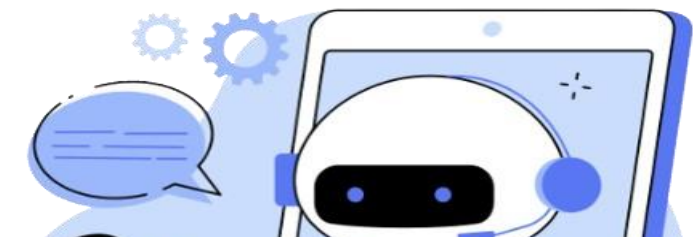


Table 7.5 Typical External Sources of Data on Which Business Analytics Can Be Based

Type of External Data	Measured By
Customer Satisfaction	• Revenue, profit • Market share, sales • Product/service survey data • Loyalty • Brand awareness • Average spend per customer
Customer Demographics	• Geographic location (distance from market) • Income level • Market size
Competition	• Market share • Competitor profitability • Advertising/promotion efforts • Preference surveys • Web chatter on products
Economic	• Population statistics • Income distribution statistics



LESSON 2.2: DATA ISSUE



Data quality can be defined as data that serves the purpose for which it is collected. It means different things for different applications, but there are some commonalities of highquality data. These qualities usually include accurately representing reality, measuring what it is supposed to measure, being timeless, and having completeness. When data is of high quality, it helps ensure competitiveness, aids customer service, and improves profitability. When data is of poor quality, it can provide information that is contradictory, leading to misguided decision-making.

LESSON 2.2: DATA ISSUE



Data privacy refers to the protection of shared data such that access is permitted only to those users for whom it is intended. It is a security issue that requires balancing the need to know with the risks of sharing too much. There are many risks in leaving unrestricted access to a company's database.

TOPIC TO BE DISCUSS:

Lesson 3: Business Analytics Technology

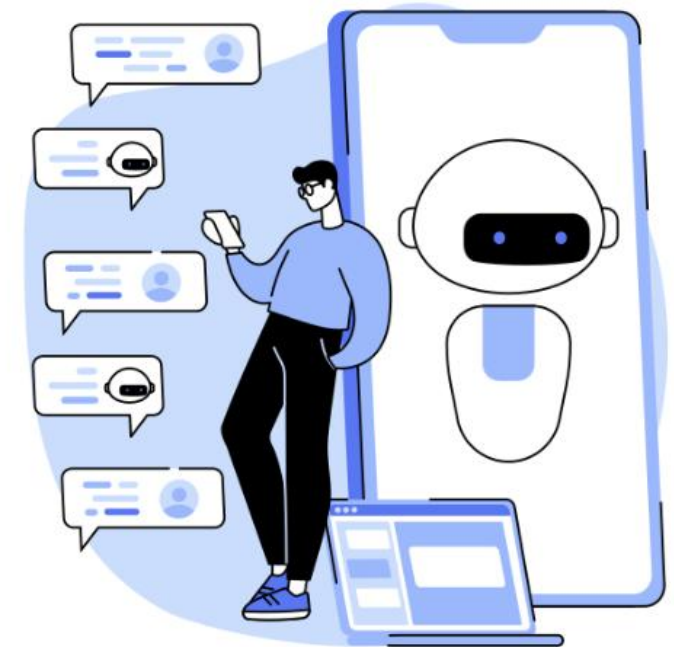


LESSON 3. BUSINESS ANALYTICS TECHNOLOGY

Table 7.6 General Information Technology (IT) Infrastructure

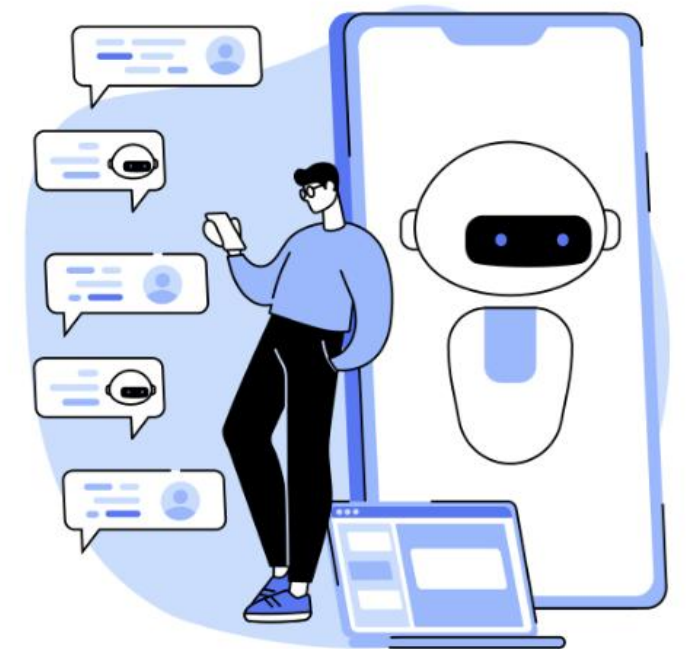
Type of Technology:

- Computer Hardware
- Computer Software
- Networking and Telecommunication Technology
- Data Management Technology



LESSON 3. BUSINESS ANALYTICS TECHNOLOGY

Database management systems (DBMS) is a data management technology software that permits firms to centralize data, manage it efficiently, and provide access to stored data by application programs. DBMS usually serves as an interface between application programs and the physical data files of structured data. DBMS makes the task of understanding where and how the data is actually stored more efficient. In addition, other DBMS systems can handle unstructured data.



LESSON 3. BUSINESS ANALYTICS TECHNOLOGY

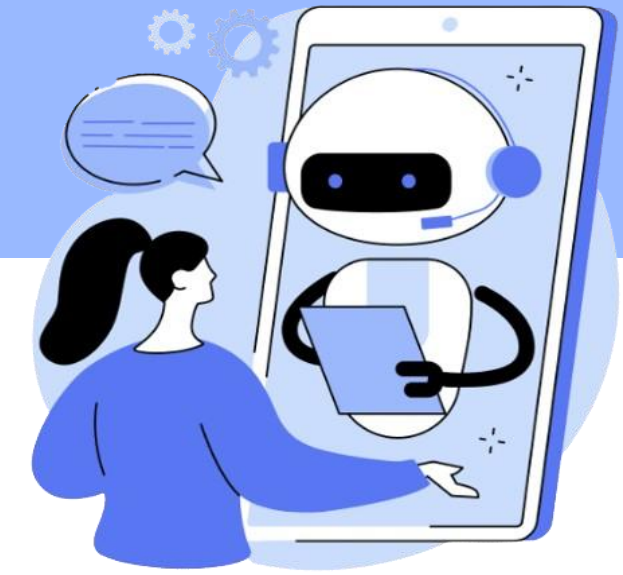


Table 7.7 Data base Encyclopedia Content

Type of Database Content Item:

- Purpose
- Time
- Source
- Scematics
- Cost
- Availability of Data
- Collection Techniques
- Collection Tools

LESSON 3. BUSINESS ANALYTICS TECHNOLOGY



Data warehouses are databases that store current and historical data of potential interest to decision makers. What a data warehouse does is make data available to anyone who needs access to it. In a data warehouse, the data is prohibited from being altered.

Data marts are focused subsets or smaller groupings within a data warehouse. Firms often build enterprise-wide data warehouses where a central data warehouse serves the entire organization and smaller, decentralized data warehouses (called data marts) are focused on a limited portion of the organization's data that is placed in a separate database for a specific population of users

LESSON 3. BUSINESS ANALYTICS TECHNOLOGY

Online analytical processing (OLAP) is software that allows users to view data in multiple dimensions.

Data mining is the application of a software, discovery-driven process that provides insights into business data by finding hidden patterns and relationships in big data or large databases and inferring rules from them to predict future behavior



LESSON 3. BUSINESS ANALYTICS TECHNOLOGY

Table 7.8 Types of Information Obtainable with Data Mining Technology

Type of Information:

- Associations
- Classification
- Clustering
- Forecasting
- Sequence



LESSON 3. BUSINESS ANALYTICS TECHNOLOGY

Text mining is a software application used to extract key elements from unstructured data sets, discover patterns and relationships in the text materials, and summarize the information.

Web mining seeks to find patterns, trends, and insights into customer behavior from users of the Web.





**THANK YOU FOR
LISTENING!**